Akhilesh Siddhanti

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SUMMARY

ML engineer at Salesforce building production retrieval, ranking, and failure-analysis systems for developer productivity. Experience spans feature engineering, model evaluation, automated retraining, versioned inference, and large-scale data processing, with measurable gains in recommendation recall and production reliability.

WORK EXPERIENCE

Salesforce — **San Francisco, CA**

Senior Member of Technical Staff — *Aug 2024 - Present*
Member of Technical Staff — *Jan 2022 - Aug 2024*
Associate Member of Technical Staff — *Jan 2021 - Jan 2022*

Selected Projects

Test Prioritizer for PRs

- Owned the TF-IDF retrieval service for PR test recommendation, scoring a 3M-test corpus per request (p95 ~5s; ~10K requests/day) for 2 downstream services
- Selected TF-IDF after benchmarking against BERT (recall@2,500 48.6% vs 20%), Word2Vec, and MiniLM, then built hourly automated retraining, versioned model hot-swap, and monitoring. Raised recall@2,500 from 33.3% to 48.6% (n=5,000 PRs) in production champion-challenger evaluation
- Built AST-based test-minimization pipelines applicable to **7% of PRs**, saving $100K compute per month
- Added normalized coverage-ranking feature (+3 pts) and XML-to-test mapping (+8 pts) features to the XGBoost model, reaching 58% recall@2,500 (gains overlap, so sub-additive).
- Cut XGBoost ranking latency by replacing ~1,000 per-test calls with a single batched prediction (score-equivalent), unblocking production rollout within 3 weeks
- Resolved OOM failures in a **566K-file similarity job** by streaming chunked cosine similarity and pruning matrix columns from **1B to 1.2M**, enabling execution within **4 GB** while preserving numerical equivalence
- *In Progress:* Developing **Agentic Test Prioritization**, exposing hand-crafted features to LLM agent as tool calls that produces 28% recall at 25 tests (comparable to 28% recall @ 700 tests on baseline)

Stacktrace Clustering

- Proposed, Owned, and built a stacktrace clustering service processing **1B+ stacktraces**, using **150+ normalization patterns** to reduce distinct stacktrace variants by approximately **98%**
- Implemented real-time embedding with **MiniLM** and indexing in **Vertex AI Vector Search**, enabling grouping and historical-fix retrieval for the Autonomous Test Fix Agent, failure-grouping capabilities for Intelligent Test Reassignment and better identification for flaky/infrastructure failures

Autonomous Test Failure Fix Agent

- Owned evaluation and experimentation for a production test-fix agent for 6,000+ engineers, collaborating with a data scientist and another SDE on architecture and development
- Built validation datasets, prompt experiments, model benchmarks, and **ICE/CodeBLEU scoring** to compare LLMs and assess generated-fix quality

Automated Visualizations: Built an LLM plugin for Claude/Cursor that generates live, database-backed **Tableau dashboards in under 5 minutes** from minimal prompts, accelerating multiple internal teams to create visualizations without requiring prior experience with Tableau

Data Insights: Owned end-to-end development of company-wide engineering dashboards and backend services, tracking changes from Work Item creation through customer deployment. Built CI pipeline health and trend monitoring with automated Slack alerts for deviations, alongside test-information and predictive-performance dashboards and a **Test Debug View** for searching and analyzing test failures.

Software Development Intern, Salesforce **May 2020 - Aug 2020**
Performed migrations from Apiary to a new Test Inventory service

SKILLS

Languages: Java, Python, C/C++, JavaScript, SQL, BigQuery, Elasticsearch, Terraform, Pandas, NumPy
Tools/Frameworks: Springboot, Kafka, REST, PyTorch, TensorFlow, Scikit-learn, Hugging Face, XGBoost, FAISS, Vertex AI, Data Lake, Git, Docker, Jenkins, Kubernetes, CI/CD, AWS, GCP, Postman
Concepts: CV, NLP, ML, LLMs, RAG, Prompt Engineering, Embedding Models, Semantic Search, Clustering

PUBLICATIONS, in International Conferences and Journals (11 total, 177 citations)

Differential fault location identification by Machine Learning **CAAI Transactions on Intelligence Technology**
A Differential Fault Attack on Plantlet (Cipher) **IEEE Transactions on Computers**

EDUCATION

MS in Computer Science, Georgia Institute of Technology, Atlanta **Aug 2019 - Dec 2020**
M.Sc. (Hons.) in Mathematics, BITS Pilani **Aug 2014 - May 2019**
B.E. (Hons.) in Computer Science, BITS Pilani **Aug 2014 - May 2019**